MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1MDCECT103 - Audio Electronics

Duration: 1.5 hour **Maximum Marks: 35**

Practical Examination – Model Question Paper

Instructions:

- **Demonstrate** the given task
- **Viva session** (Minimum 5 questions from the Practical Module)
- **Lab Report**

Evaluation Criteria:

- **Viva**: 10 marks
- **Lab Report**: 10 marks
- **Demonstration**: 15 marks

Tasks: *(Choose any one)*

1. Create a conversion cable from XLR to TRS and test its functionality with microphones and mixers.
2. Design and assemble an adapter to connect TRS audio outputs to RCA inputs for audio devices.
3. Build a simple adapter to connect RCA outputs to a 3.5mm headphone jack.
4. Create an adapter that converts a mono signal to stereo and test it with various audio sources.
5. Construct a dual 3.5mm to dual RCA cable for connecting audio devices to mixers or amplifiers.
6. Design and construct a basic audio amplifier using transistors or op-amps and test its performance with a speaker.
7. Design and construct a two-channel mixer using op-amps or transistors.
8. Construct a low-pass filter circuit to observe its audio signals and frequency response.
9. Design and implement a tone control circuit (bass, midrange, treble) and test its effect on audio playback.
10. Create a noise filter circuit to reduce unwanted noise from audio signals.
11. Build a subwoofer amplifier circuit and test its performance with low-frequency audio signals.
12. Make an IC UM66-based signal injector circuit.
13. Make an electronic piano using NE555 IC